Yuyang Cheng

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EDUCATION

Sichuan University *Bachelor of Cybersecurity* *Sep 2022 – Jun 2026*

- **GPA:** 3.65/4.0
- **IELTS:** 7.0

RESEARCH EXPERIENCE

The Hong Kong University of Science and Technology (Guangzhou) *Guangzhou, China*
Research Assistant *advised by* Prof. Menglin Yang *in* DIGAI Lab *Jun 2026 – Present*
Personalized Agents; AI Security.

- Identified a scene-grounded safety vulnerability in multimodal large language models. [P.1]
- Leading a comprehensive survey of agent workflows.
- Investigating latent reasoning in LLM-based dense retrievers.

The Hong Kong Polytechnic University *Hong Kong, China*
Research Intern (Remote) *advised by* Prof. Songlin Xu *Apr 2026 – Present*
Human-AI Alignment

- investigating misalignment in how human-written and LLM-generated peer reviews express skepticism.

Incoming Research Assistant *Dec 2026 (expected)*
Planned Research: Closed-Loop Human-AI Integration for Cognitive and Physical Augmentation

West China Biomedical Big Data Center, West China Hospital *Chengdu, China*
Research Intern *advised by* Prof. Wei Zhang *and* Prof. Kang Li *May 2025 – Mar 2026*
Mental Health Knowledge Graphs.

- Collaborated with clinicians and engineers to build a mental health knowledge graph for outpatient consultations.

Sichuan University *Chengdu, China*
Undergraduate Researcher *advised by* Prof. Yong Zhao *Sep 2024 – Mar 2026*
Agent Workflows; Recommender Systems; Blockchain.

- Designed a hierarchical workflow framework for multi-agent collaboration. [C.1]
- Developed a collaborative filtering algorithm for personalized recommendation. [J.1]
- Developed security and efficiency optimizations for account-based anonymous blockchain systems. [C.2, J.2]

PUBLICATIONS (* Equal contribution)

Preprints

[P.1] **Scenes as an Attack Surface: Uncovering Scene-Grounded Safety Failures in MLLMs**
Tongxi Wu*, **Yuyang Cheng***, Jiacheng Xian*, Hu Huang, Zhihao Wu, Menglin Yang, Jian Zhang, Yang Gao
Preprint 2026

[P.2] **CreAgentive: An Agent Workflow Driven Multi-Category Creative Generation Engine**
Yuyang Cheng*, Linyue Cai*, Changwei Peng, Yumiao Xu, Rongfang Bie, Yong Zhao
Preprint 2025

Journal Articles

[J.1] **OTML-MF: A Collaborative Filtering Recommendation Algorithm Based on Multi-dimensional Feature Fusion and Improved LSH**
Yuyang Cheng*, Yichen Tan*, Zixuan Zhou, Yong Zhao
Information Sciences 2026

[J.2] **The Optimization of Batch Processing and Micro-Payment Systems in Account-Based Anonymous Blockchain Systems**
Yichen Tan*, **Yuyang Cheng***, Lu Ding, Yong Zhao
Blockchain: Research and Applications 2026

Conference Papers

[C.1] **HAWK: A Hierarchical Workflow Framework for Multi-Agent Collaboration**
Yuyang Cheng*, Yumiao Xu*, Chaojia Yu, Yong Zhao
AgentIR Workshop @ SIGIR 2025

[C.2] **Multi-signature Authentication and Simulation Extractability Security Optimization for Account-based Blockchain Anonymous Systems**
Yichen Tan*, **Yuyang Cheng***, Lu Ding, Yong Zhao
International Conference on Computer Application and Information Security (ICCAIS 2024)

SKILLS

- **Programming:** Python, C/C++, Java, SQL, Neo4j, Git, Docker, Linux, LaTeX
- **Frameworks:** PyTorch, Hugging Face, LangChain, LangGraph, AutoGen

AWARDS & HONORS

- Outstanding Graduate – Sichuan University *2026*
- Finalist, Mathematical Contest in Modeling *2025*
- First-Class Scholarship – Sichuan University *2023, 2024*
- Outstanding Student – Sichuan University *2023*